

- JOSÉ LUIS CASTIGLIONI AND RODOLFO C. ERTOLA-BIRABEN, *Rough sets semantics for three-valued Priest's da Costa logic*.

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Rough sets were introduced by Pawlak and his co-workers in the early 1980s (for instance, see [2] and [3]). In [1] we studied a three-valued first-order logic which may be given in the signature $\{\wedge, \vee, \neg, D\}$. This logic may also be presented as a modal logic. In this talk we present a rough sets semantics for the mentioned logics.

Given a first-order language L , which, for simplicity, we assume only with a non-empty set of n -ary predicate letters, we define an L -structure as a non-empty set U (its universe) together with a partition of U . This partition of U , suitably defines one on U^n , for every $n \in \mathbb{N}$. We associate an n -ary relation $R^U \subseteq U^n$ to every n -ary predicate letter R in L . Note that each $R^U \subseteq U^n$ may be viewed as a rough set on U^n . As usual, we indicate the upper approximation of R^U by $\overline{R^U}$, and its lower approximation by $\underline{R^U}$. An interpretation of a language L is a pair $\mathcal{I} = (\mathcal{A}, e)$, where $\mathcal{A}$ is an L -structure and $e : \text{Var}_L \rightarrow U$ is a function assigning an element of U to each variable of L .

We define the notion of semantic consequence as for classical logic, that is to say, given $\Gamma \cup \{\alpha\} \subseteq \mathfrak{F}_L$, we write $\Gamma \models \alpha$ if for every interpretation $\mathcal{I}$ of L , it holds that $\min\{v_{\mathcal{I}}(\gamma)\} \leq v_{\mathcal{I}}(\alpha)$. Here, the recursively defined function $v_{\mathcal{I}} : \mathfrak{F}_L \rightarrow \mathbf{3}$ is the three-valuation associated to the interpretation $\mathcal{I}$.

We prove that the first-order logic considered is sound and complete relatively to the rough sets semantics described above.

[1] JOSÉ LUIS CASTIGLIONI AND RODOLFO C. ERTOLA-BIRABEN, *Modalities combining two negations*, **Journal of Logic and Computation**, vol. 35 (2025), no. 3, exae020.

[2] ZDZISŁAW PAWLAK, *Rough Sets*, **International Journal of Computer and Information Sciences**, vol. 11 (1982), pp. 341–356.

[3] ZDZISŁAW PAWLAK, *Rough Sets. Theoretical Aspects of Reasoning about Data*, Series D: System Theory, Knowledge Engineering and Problem Solving, Kluwer Academic Publishers, 1991.